

Financial Fraud Detection Dashboard

Comprehensive Project Documentation

Project Type	Data Science & ML - Capstone Project
Technologies	Python, Streamlit, SQLite, scikit-learn, XGBoost, LightGBM, TensorFlow
Dataset Size	100,065 transactions 8,000 unique customers 1.80% fraud rate
Models Deployed	7 ML algorithms (Supervised + Unsupervised + Deep Learning)
Dashboard Type	Interactive Streamlit web app with 5 feature-rich tabs

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1. Executive Summary

This Financial Fraud Detection Dashboard is a capstone data science project that combines Machine Learning, data engineering, and interactive visualization to detect and monitor fraudulent transactions in real-time.

Project Goals

- Build a scalable fraud detection system using 7 different ML algorithms
- Create an interactive Streamlit dashboard for fraud monitoring and investigation
- Engineer realistic synthetic transaction data with 5 distinct fraud patterns
- Compare supervised vs unsupervised vs deep learning approaches
- Provide actionable insights to fraud investigation teams

Key Results

- XGBoost & LightGBM: 99.5%+ accuracy, 94% fraud recall (best performers)
- ROC-AUC score: 0.999 (excellent discrimination between fraud/legitimate)
- All 7 models evaluated: supervised models outperform unsupervised by ~25%
- Dashboard with 5 interactive tabs for EDA, model comparison, and investigation

2. System Architecture & Workflow

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The fraud detection system follows a complete data science pipeline:

STEP 1: DATA GENERATION

- └ Synthetic dataset generator (100,065 transactions)
- └ 8,000 unique customers
- └ 5 engineered fraud patterns (Card Testing, Account Takeover, Geo-Velocity, etc.)
- └ Full year of transaction history (2025-01-01 to 2025-12-31)

STEP 2: ETL PIPELINE

- └ Extract: CSV file loaded into Python
- └ Transform: Data cleaning, normalization, feature engineering
- └ Load: SQLite database (fraud_detection.db)
- └ Indexing: Created on customer_id, is_fraud, timestamp for fast queries

STEP 3: FEATURE ENGINEERING

- └ Temporal features: hour of day, day of week, month, weekend flag
- └ Customer behavior: time since last transaction, amount-to-avg-spend ratio
- └ Geographic: distance from home city, location changes
- └ Device & merchant: new device flag, merchant risk category
- └ Total: 24 engineered features for ML models

STEP 4: MODEL TRAINING

- └ 7 models trained on 75% of data (stratified split)
- └ Evaluation on held-out 25% test set
- └ Metrics tracked: Accuracy, Precision, Recall, F1, ROC-AUC, Avg Precision
- └ Models saved to artifacts/ folder for instant dashboard loading

STEP 5: INTERACTIVE DASHBOARD

- └ Streamlit web app (Python-based)
- └ 5 tabs: Overview, EDA, Model Comparison, Transaction Investigator, ETL Pipeline
- └ Real-time model selection and transaction scoring
- └ Deployed on local machine (<http://localhost:8501>)

3. Dataset Overview

The project uses a synthetically generated dataset that mimics real-world banking transactions with engineered fraud patterns based on actual fraud research.

Dataset Statistics

Total Transactions	100,065
Unique Customers	8,000
Transaction Volume	₹2.3 Billion
Fraudulent Transactions	1,801 (1.80%)
Fraud Exposure	₹49 Million (2.1% of total volume)
Time Period	Full Year 2025
Features per Transaction	24 engineered features
Merchant Categories	15 (Grocery, Electronics, Crypto, Gambling, etc.)

Fraud Patterns Engineered

Card Testing: Rapid small-value transactions on newly compromised cards. Avg 4-7 txns in 15 minutes. (420 cases)

Account Takeover: High-value spending bursts after dormancy on a compromised account. (360 cases)

Geo-Velocity Mismatch: Large transactions in distant cities within impossible timeframes. (360 cases)

Odd-Hour High-Value: Unusual spending at 12am-4am, high-risk merchants (Crypto, Gambling). (360 cases)

New Device + High-Risk: First-time device combined with high-risk merchant category. (301 cases)

4. ETL Pipeline

Extract

Raw synthetic transaction data is generated as fraud_transactions.csv. This CSV contains 100,065 rows with initial features and fraud labels.

Transform

- Missing value handling: Numeric columns imputed with median; categorical with 'Unknown'
- Data validation: Remove duplicate transaction IDs
- Normalization: Min-max scaling applied to transaction amounts
- Feature engineering: Create derived features (hour, day of week, distance, ratios)
- Type casting: Boolean fields converted to integers for SQLite compatibility

Load

Cleaned data is loaded into SQLite database (fraud_detection.db) with indexes on customer_id, is_fraud, and timestamp for optimized query performance.

Database Schema

transaction_id	PRIMARY KEY - Unique identifier
customer_id	Foreign key to customer profile
timestamp	Transaction date/time (YYYY-MM-DD HH:MM:SS)
amount	Transaction amount in Rupees
merchant_category	Category of merchant (Grocery, Electronics, etc.)
is_fraud	Binary label (0=legitimate, 1=fraudulent)

5. Feature Engineering

24 features are engineered from raw transaction data to help ML models learn fraud patterns. These features capture temporal, behavioral, geographic, and merchant-level signals.

Feature Categories

Temporal Features

- `hour_of_day`: Transaction hour (0-23)
- `day_of_week`: Day name (Monday-Sunday)
- `is_weekend`: Binary flag for weekend
- `month`: Month name (January-December)

Behavioral Features

- `minutes_since_last_txn`: Time gap from previous transaction (minutes)
- `amount_to_avg_spend_ratio`: Spending relative to customer average
- `account_age_days`: Customer account age

Geographic Features

- `distance_from_home_km`: Haversine distance to home city
- `city`, `latitude`, `longitude`: Transaction location

Device & Merchant Features

- `is_new_device`: First-time device usage flag
- `device_type`: POS Terminal, Mobile, Desktop, ATM, Tablet
- `merchant_category`: 15 risk-weighted categories
- `channel`: Transaction channel (POS, Online, ATM, Mobile App, Recurring)

Customer Profile Features

- `customer_risk_segment`: Low/Medium/High risk customer class
- `card_type`: Visa, Mastercard, RuPay, Amex

6. Machine Learning Models

Seven machine learning models are trained and compared to detect fraud. They fall into three categories: Supervised (learn from labeled data), Unsupervised (detect outliers), and Deep Learning (neural networks).

● Logistic Regression

Type: Supervised (Linear)

Description: Fast baseline model that learns linear decision boundaries.

Performance: 93.46% Accuracy | 0.984 ROC-AUC

Best Used For: Interpretability, quick predictions

● Random Forest

Type: Supervised (Tree Ensemble)

Description: Ensemble of decision trees. Robust to outliers and handles non-linearity.

Performance: 99.47% Accuracy | 0.998 ROC-AUC

Best Used For: Balance of accuracy and interpretability

● XGBoost

Type: Supervised (Gradient Boosting)

Description: Sequential tree boosting with regularization. State-of-the-art performance.

Performance: 99.52% Accuracy | 0.999 ROC-AUC

Best Used For: Best accuracy, handles imbalanced data well

● LightGBM

Type: Supervised (Gradient Boosting)

Description: Fast gradient boosting framework, optimal for large datasets.

Performance: 99.51% Accuracy | 0.999 ROC-AUC

Best Used For: Speed + accuracy, production deployments

● Isolation Forest

Type: Unsupervised (Anomaly Detection)

Description: Detects outliers by isolating anomalies. No fraud labels needed.

Performance: 97.97% Accuracy | 0.947 ROC-AUC

Best Used For: Detecting novel/unknown fraud patterns

One-Class SVM

Type: Unsupervised (Support Vector)

Description: Learns boundary of normal transactions, flags anything outside.

Performance: 96.25% Accuracy | 0.858 ROC-AUC

Best Used For: Highly anomalous transactions

Autoencoder

Type: Deep Learning (Neural Network)

Description: Neural network trained to reconstruct normal transactions. Flags high reconstruction error.

Performance: 94.11% Accuracy | 0.783 ROC-AUC

Best Used For: Complex non-linear patterns

Key Insights

- Supervised models (XGBoost, LightGBM, RF) achieve 99%+ accuracy because they learn exact fraud patterns
- Unsupervised models (Isolation Forest, One-Class SVM) catch novel fraud types not seen in training
- Deep learning (Autoencoder) adds complexity but doesn't improve performance on this dataset
- Best strategy: Use supervised models for speed, combine with unsupervised for coverage of unknown frauds

7. Dashboard Components & Tabs

The Streamlit dashboard provides 5 interactive tabs for different aspects of fraud monitoring and analysis.



Overview Tab

Executive summary with KPIs and trend analysis

Key Features

- Total transactions, fraud count, fraud rate at a glance
- Transaction volume trend over time (daily rolling)
- Fraud pattern pie chart (5 types)
- Geographic heatmap of fraud by city
- Peak fraud hours analysis



Exploratory Analysis Tab

Deep-dive into patterns and relationships

Key Features

- Amount distribution: Fraud vs Legitimate (fraud is 2-10x higher)
- Distance from home city (fraud transactions often far away)
- New device usage patterns (strong fraud indicator)
- Merchant category breakdown
- Correlation heatmap of all features



Model Comparison Tab

Performance metrics and model evaluation

Key Features

- Side-by-side model metrics table (Accuracy, Precision, Recall, F1, ROC-AUC)
- ROC & Precision-Recall curves for all 7 models
- Feature importance ranking (XGBoost top features)
- Confusion matrices for each model
- Training time and model size comparison



Transaction Investigator Tab

Real-time fraud investigation tool

Key Features

- Alert feed ranked by fraud risk score (0-100)
- Select model for scoring (any of 7 models)
- Transaction details: amount, location, time, device, merchant
- Confirmed fraud labels for model evaluation
- Drill-down into individual transactions

ETL & Pipeline Tab

Data pipeline documentation

Key Features

- ETL process overview and statistics
- Database schema description
- Sample of cleaned data from SQLite
- Feature list with descriptions
- Data quality metrics

8. Key Findings & Insights

Finding 1: Fraud is Rare but High-Impact

Only 1.8% of transactions are fraudulent, but they account for 2.1% of total volume. This class imbalance is handled using stratified train-test splits and class weighting in models.

Finding 2: Amount is a Strong Signal

Fraudulent transactions average ₹27,000 vs legitimate average ₹8,100 (3.3x higher). Simple rule-based detection on high amounts catches ~40% of fraud with minimal false positives.

Finding 3: Geographic Anomalies Matter

Transactions far from home city (>500km) have 5x higher fraud rate. Geo-velocity mismatches (distant cities, short timeframes) are strong fraud indicators.

Finding 4: New Device + High-Risk Merchant = Alert

First-time device + Crypto/Gambling/Money Transfer combo has 12% fraud rate (vs 1.8% overall). Simple rule: flag these combinations for manual review.

Finding 5: Supervised >> Unsupervised

XGBoost and LightGBM achieve 99.5% accuracy because they learn exact fraud patterns. Unsupervised methods (Isolation Forest) only achieve 97% by flagging generic outliers. Best approach: Use supervised for known frauds, unsupervised for novel patterns.

9. User Guide & How to Use

Installation & Setup

1. Extract the fraud_detection_project.zip file
2. Open Command Prompt in the extracted folder
3. Run: `python -m pip install -r requirements.txt` (wait 1-2 minutes)
4. Double-click RUN_DASHBOARD.bat to launch

Dashboard Workflow

Follow this workflow to investigate fraud:

Start in Overview Tab: Get overall fraud statistics and trends. Check if fraud is rising or falling.

Explore Data Patterns (EDA Tab): Understand which features correlate with fraud (amount, distance, device).

Compare Models (Model Comparison Tab): Review which model performs best. Check feature importance.

Investigate Alerts (Transaction Investigator Tab): Use the best-performing model to score transactions. Review high-risk transactions.

Review Pipeline (ETL Tab): Understand how data flows from CSV → SQLite → ML models → Dashboard.

How to Use Each Tab

- Overview: Monitor KPIs, check fraud rate trends, identify peak fraud hours.
- EDA: Drill into feature distributions, find patterns in fraudulent transactions.
- Model Comparison: Compare all 7 models side-by-side. Choose best model for your use case.
- Transaction Investigator: Select model → see risk scores for transactions → investigate high-risk ones
- ETL Pipeline: Understand data flow, database schema, feature engineering logic.

10. Troubleshooting & FAQ

Q: Dashboard won't launch / 'Streamlit not found' error

A: Make sure you ran: `python -m pip install -r requirements.txt` first. If using Anaconda, use the full path: `C:\Users\YourName\anaconda3\python.exe -m streamlit run app.py`

Q: Port 8501 already in use

A: Another Streamlit instance is running. Either close it, or edit `RUN_DASHBOARD.bat` to use a different port (change 8501 to 8502)

Q: Which model should I use for production?

A: XGBoost or LightGBM. Both achieve 99.5% accuracy and ROC-AUC 0.999. LightGBM is faster for real-time scoring.

Q: How do I interpret the risk score (0-100)?

A: Score = normalized fraud probability from the selected model $\times 100$. Higher = more likely fraud. Typical threshold: flag anything >50 for review.

Q: Can I update the models with new transactions?

A: Yes. Run `train_models.py` after updating `fraud_transactions.csv` and running `etl_pipeline.py`. Models will retrain and artifacts/ will update.

Q: Why does unsupervised (Isolation Forest) have lower accuracy?

A: Unsupervised models don't use fraud labels—they only flag outliers. They excel at catching novel fraud types not seen in training, but have lower accuracy on known patterns.

Q: How is this dataset synthetic, not real?

A: Dataset is synthetically generated to contain exact fraud patterns, making it ideal for model evaluation and demo purposes. Real data has messier patterns.

Quick Reference Guide

Save this page for quick lookup during investigations:

Total Dataset	100,065 transactions 8,000 customers 1.8% fraud rate
Top Model	XGBoost: 99.52% accuracy, 94.9% recall, 0.999 ROC-AUC
Fraud Pattern	Card Testing (420), Account Takeover (360), Geo-Velocity (360), Odd-Hour (360), New Device (301)
Feature Count	24 engineered features (temporal, behavioral, geographic, merchant, device)
Dashboard Tabs	Overview, EDA, Model Comparison, Transaction Investigator, ETL Pipeline
Launch Command	Double-click RUN_DASHBOARD.bat or run: python -m streamlit run app.py
Browser URL	http://localhost:8501